

## **Social Media and Negative Personality Traits**

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### **Abstract**

Social media greatly influences human behavior and the way people interact with others via a virtual space. Some studies have sought to examine the relationships between virtual perceptions and abnormal, often dark, personalities. Participants were given surveys that assessed the presence of narcissism, machiavellianism, and psychopathy (NMP) as well as time they spend on social networking sites, number of photos they have posted, and number of selfies they have posted. Participants had higher levels of NMP when they spent longer on social media, posted more photos, and posted more selfies. These findings suggest that there is an association between interaction with social media and personality traits that are centered around grandiose attention seeking and manipulation.

### **Social Media and Negative Personality Traits**

The influence of social media on the human condition takes many forms. Exposure to social media can have varying effects on behavior and personality traits. In particular, personality traits concerning grandiosity, morality, and manipulation. Many researchers have studied the relationship between social media and certain dark cognitions and behaviorisms. Narcissism, Machiavellianism, and Psychopathy are at the forefront of this dispute.

For example, Singh and Donahue (2018) found that individuals with narcissistic traits are more active and spend more time on social networks. Participants completed a series of questionnaires assessing narcissistic tendencies and social media behaviors. The researchers found that the main motivation behind the association between narcissism and social media was admiration. Those with higher levels of narcissism used social media with self-interest intentions. In other words, these individuals posted pictures to elicit ego-stroking attention from others, rather than for social networking purposes. However, this study lacks introspection on other negative leaning personality traits.

Some studies have worked to bridge this gap between varying negative traits. According to McCain et al. (2016), there was also a positive association between selfie taking and narcissism. Moreover, the researchers found that there were similar dimensions of social media behavior between narcissism and both psychopathy and machiavellianism. Therefore, there is a proposed connection amongst these darker personalities and how they spend their time virtually. This begs the question of whether those with these darker traits act similarly online consistently, or if at some point they differ.

A few studies have examined the often minute differences between narcissism, machiavellianism, and psychopathy. Fox and Rooney (2015) researched the ties between these

three negative traits but with a focus on self-objectification in men. The results revealed that there was an increase in likelihood of not only selfie posting, but selfie editing for subjects high in narcissism but not psychopathy. However, neither selfie posting nor selfie editing were associated with high rates in machiavellianism. It seems that there was a split in behavior between the three once selfies and photos in general were accounted for.

There seems to be some tensions amongst the association between grandiose or immoral traits and social media, along with how similarly people with different darker traits act. In this study I will examine the existence of a correlation between negative personality traits and interactions with social media. More specifically, this study looks into the relationship between rates of narcissism, machiavellianism, and psychopathy and varying levels of interaction with social media. Interaction will be operationalized as minutes spent on social media, number of photos posted, and number of selfies posted. I expect that there will be higher averages for social media interaction as the presence of negative personality traits increase. Specifically, I expect that subjects with higher levels of narcissism, machiavellianism, and psychopathy will spend longer on social media, post more pictures, and post more selfies. Additionally, I expect that there will be higher rates of narcissism in participants in comparison to machiavellianism and psychopathy.

## **Methods**

The study was conducted using surveys and provided by Ethan Hurwitz. The final sample included 800 subjects. The mean age was 29.5 years old with a range from 18 years old to 40. All of the observations are from the questionnaire results. There are six variables in this study. All of the variables are rate of narcissistic traits, rate of machiavellian traits, rate of psychopathy, amount of time spent on social media in minutes, number of selfies posted, and number of photos

posted. A high score in Narcissism, Machiavellianism, and Psychopathy is considered a negative outcome. In order to investigate the relationship between dark personality traits and social media, I first looked into time spent on social media. RStudio was used to find the average rates of narcissism, machiavellianism, and psychopathy for different levels of time spent on social media. Numerical summaries for both zero minutes spent on social media and any amount of time greater than zero minutes were created. The same was done for the amount of photos posted and the amount of selfies posted. Age demographics were also discovered this way. This information will reveal if there is an effect between the three negative personality traits and the three social media behaviors. Nine visualizations were created using RStudio. Three scatterplots were created to assess relationships between minutes spent on social media and narcissism, machiavellianism, and psychopath. Six bar graphs were created to investigate associations between the three negative personality traits and amount of photos and selfies posted on social media respectively.

## **Results**

The computed summary results for mean narcissism, machiavellianism, and psychopathy (NMP) based on different amounts of time spent on social media revealed an increase in means when time was not zero. The overall averages for NMP, despite social media interaction, were 2.97, 2.42, and 2.54 respectively (see Figure 1). The means for NMP were 2.70, 2.29, and 2.47 respectively when minutes on social media was zero, and they were 3.04, 2.46, and 2.56 when minutes on social media were more than zero (see Figures 2 and 3). For the number of photos posted, the means for NMP were 2.93, 2.35, and 2.50 when no pictures were posted, and were 3.02, 2.50, and 2.59 when pictures were posted (see Figures 4 and 5). The NMP means when selfies were not posted were 2.86, 2.35, and 2.48, while the means were 3.13, 2.54, and 2.63

when selfies were posted (see Figures 6 and 7). For all of the computed summaries, the mean rate of narcissism was overall larger than both machiavellianism and psychopathy. The mean rate of machiavellianism was lower than psychopathy for every summary as well. Three graphs revealed a positive correlation between NMP and time spent on social media (see Figures 8, 9, and 10). All graphs concerning the relationship between NMP and either number of photos posted or number of selfies posted revealed a positive correlation (see Figures 11, 12, 13, 14, 15, and 16). We know this because the line of best fit on all nine graphs has a gradual upward trend.

Figure 1

```
A tibble: 1 × 3
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  <dbl> <dbl> <dbl>
1  2.42  2.97  2.54
```

Figure 2

```
A tibble: 1 × 3
  narc_mean mach_mean psycho_mean
  <dbl> <dbl> <dbl>
1  2.70  2.29  2.47
```

Figure 4

```
A tibble: 1 × 3
  narc_mean mach_mean psycho_mean
  <dbl> <dbl> <dbl>
1  2.93  2.35  2.50
```

Figure 3

```
A tibble: 1 × 3
  narc_mean mach_mean psycho_mean
  <dbl> <dbl> <dbl>
1  3.04  2.46  2.56
```

Figure 5

```
A tibble: 1 × 3
  narc_mean mach_mean psycho_mean
  <dbl> <dbl> <dbl>
1  3.02  2.50  2.59
```

Figure 6

```
# A tibble: 1 × 3
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  <dbl>      <dbl>      <dbl>
1    2.86      2.35      2.48
```

Figure 7

```
A tibble: 1 × 3
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  <dbl>      <dbl>      <dbl>
1    3.13      2.54      2.63
```

Figure 8

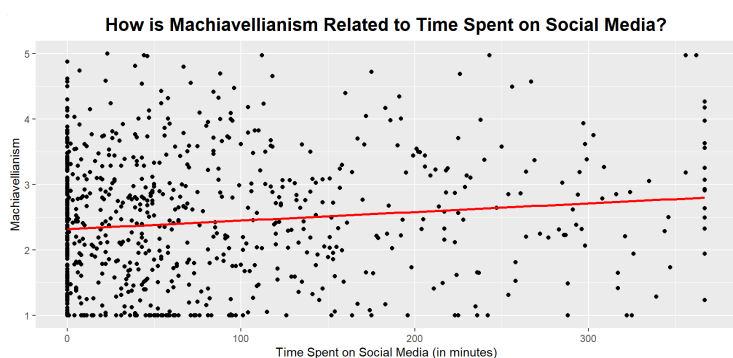


Figure 9

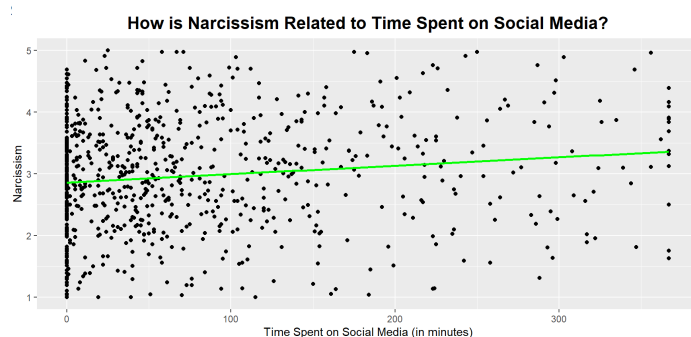


Figure 10

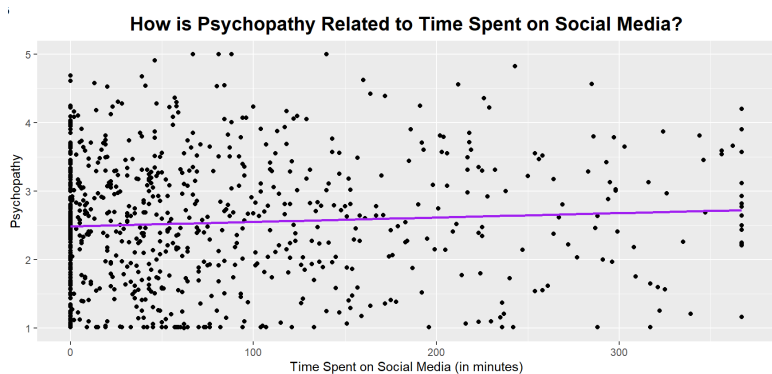


Figure 11

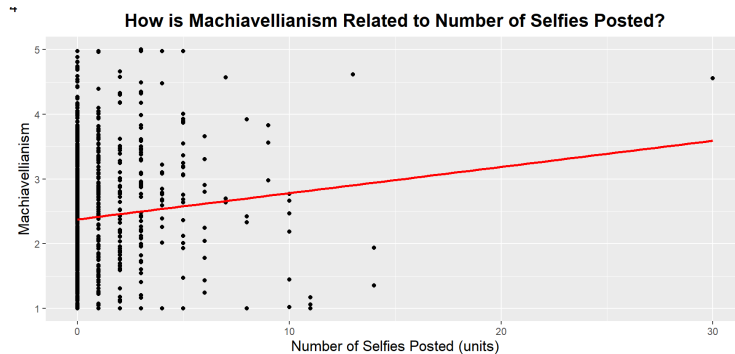


Figure 12

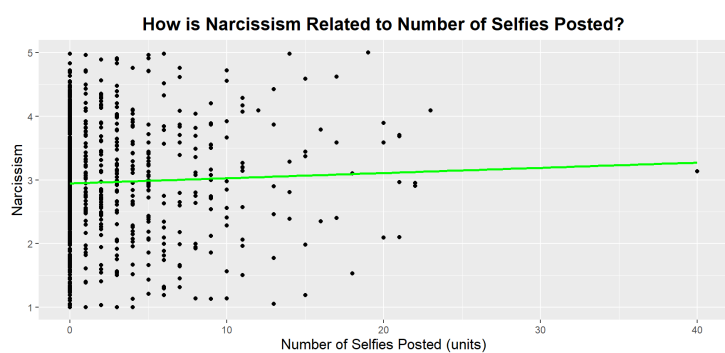


Figure 13

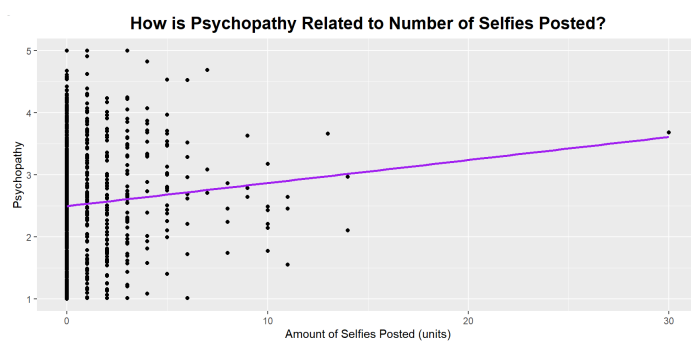


Figure 14

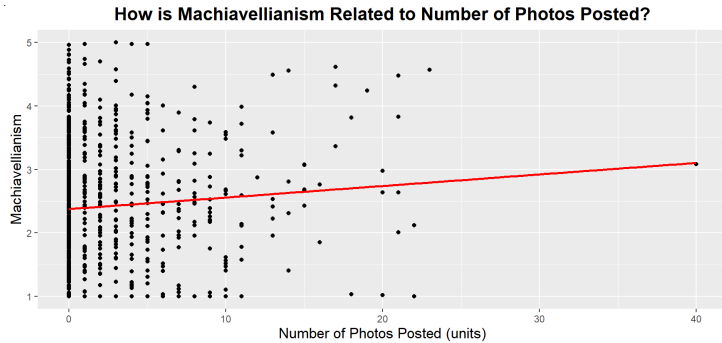


Figure 15

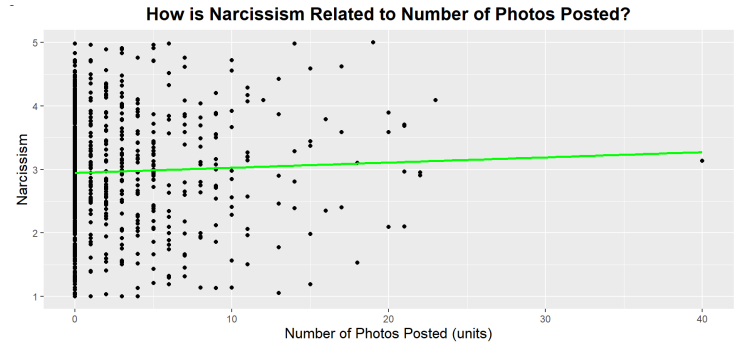
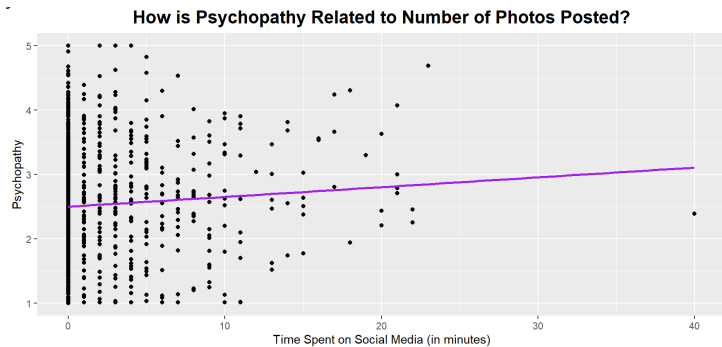


Figure 16



## Discussion

There was an increase in minutes spent on social media, selfies posted, and photos posted as the rates of narcissism, machiavellianism, and psychopathy increased. Moreover, the results indicated that the subjects on average had higher scores of narcissism, followed by psychopathy, and then machiavellianism. These findings support my hypothesis that there will be an increase in dark personality traits as social media interaction increases. In other words, individuals with higher levels of grandiosity, immorality, and cunningness may spend more time utilizing virtual spaces to both socialize and gain the attention of others. The reason for this is unclear as an



outside variable, such as the need for others to like you, could be a contributing factor to this association.

A major limitation of this study is the lack of insight on gender. It is possible that there is a significant difference between how men, women, and other genders use social media. Additionally, which of the NMP traits subjects score higher in could differ based on gender as well. Furthermore, the nature of this study could elude a problem of environment for the subjects. The area the participants were in could have affected how they responded to the surveys. Loud environments could have increased the presentation of negative personality traits and vice versa for quieter environments. Likewise, participants might have been unintentionally underrepresenting themselves as one of the NMP traits for the sake of looking good. This could be especially true given the nature of the study. Future research should take into account gender when surveying subjects and should control for the environment. Researchers can do this by having participants come into their lab to take the survey. This would reduce the problem of varying environments as well as the perception of being viewed by others. The researchers can ensure that the participants are not watched by anyone when they are taking their surveys, and do not therefore feel like they need to lie.

The results of this study reveal a tie between participation in virtual spaces and the presence of less socially accepted and more morally ambiguous personalities. Social media has made connecting with other people easier in a sense. It has become an extremely common practice to make friends online and curate a virtual and public image for others to witness. It is this inherent desire to be viewed by others, which is encouraged through social media, that might elicit certain behavioral responses. Likewise, individuals who seek to manipulate others' perception of them, or gain the approval of others may flock to these sites. It is perhaps the

environment that social media creates that should be studied more extensively, knowing that it is potentially used more heavily by individuals with more negative traits.

## References

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